

```
In [ ]: import torch
        torch.cuda.empty_cache()
        torch.cuda.ipc_collect()
```

```
In [ ]: import os
        os.environ["PYTORCH_CUDA_ALLOC_CONF"] = "expandable_segments:True"
```

```
In [ ]: !pip install torch torchvision torchaudio --index-url https://download.pytorch.o
```

```
In [ ]: !pip install transformers datasets peft bitsandbytes accelerate scikit-learn -U
```

```
In [1]: import torch

        print("CUDA available:", torch.cuda.is_available())

        if torch.cuda.is_available():
            print("GPU name:", torch.cuda.get_device_name(0))
```

CUDA available: True

GPU name: NVIDIA RTX A6000

```
In [2]: # =====
        # IMPORTS
        # =====

        import pandas as pd
        import numpy as np
        import gc
        import torch
        import torch.nn as nn

        from sklearn.model_selection import train_test_split
        from sklearn.metrics import (
            accuracy_score,
            f1_score,
            classification_report,
            confusion_matrix
        )

        from sklearn.utils.class_weight import compute_class_weight

        from datasets import Dataset

        from transformers import (
            AutoTokenizer,
            AutoModelForCausalLM,
            TrainingArguments,
            Trainer, BitsAndBytesConfig
        )

        from peft import prepare_model_for_kbit_training, LoraConfig, get_peft_model
```

```
/opt/conda/lib/python3.10/site-packages/tqdm/auto.py:21: TqdmWarning: IPProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
```

```

In [3]: # =====
# LOAD BALANCED DATASETS
# =====

course_balanced = pd.read_csv(
    "course_balanced-new.csv"
)

teacher_balanced = pd.read_csv(
    "teacher_balanced-new.csv"
)

university_balanced = pd.read_csv(
    "university_balanced-new.csv"
)

# =====
# LOAD SHARED FIXED TEST SETS
# =====

teacher_test = pd.read_csv(
    "teacher_test.csv"
)

course_test = pd.read_csv(
    "course_test.csv"
)

university_test = pd.read_csv(
    "university_test.csv"
)

```

```

In [4]: train_df = pd.concat(
    [university_balanced, course_balanced]
).reset_index(drop=True)

test_df = teacher_test.reset_index(drop=True)

print("Train:", train_df.shape)
print("Test:", test_df.shape)

```

Train: (8436, 5)

Test: (844, 7)

```

In [5]: # =====
# TRAIN / VALIDATION SPLIT
# =====

train_df, val_df = train_test_split(
    train_df,
    test_size=0.1111,
    stratify=train_df["sentiment"],
    random_state=42
)

```

```

train_df = train_df.reset_index(drop=True)
val_df = val_df.reset_index(drop=True)

# FIXED SHARED TEST SET
test_df = test_df.reset_index(drop=True)

print("Train:", train_df.shape)
print("Validation:", val_df.shape)
print("Test:", test_df.shape)

```

Train: (7498, 5)
 Validation: (938, 5)
 Test: (844, 7)

```

In [6]: # =====
# LOGIN
# =====
from huggingface_hub import login
hf_token = input("Enter your Hugging Face token: ")
login(hf_token)

```

```

In [7]: # =====
# LABEL MAPPING
# =====

label2id = {
    "negative": 0,
    "neutral": 1,
    "positive": 2
}

id2label = {
    0: "negative",
    1: "neutral",
    2: "positive"
}

train_df["label"] = train_df["sentiment"].map(label2id)
val_df["label"] = val_df["sentiment"].map(label2id)
test_df["label"] = test_df["sentiment"].map(label2id)

```

```

In [8]: # =====
# CLASS WEIGHTS
# =====

classes = np.array([0, 1, 2])

class_weights = compute_class_weight(

    class_weight="balanced",

    classes=classes,

    y=train_df["label"].values
)

class_weights = torch.tensor(
    class_weights,
    dtype=torch.float
)

```

```
)

print("\nClass Weights:")
print(class_weights)
```

Class Weights:
 tensor([1.3308, 3.1086, 0.5190])

```
In [9]: # =====
# FORMAT INSTRUCTION
# =====

def format_instruction(row):

    return f"""<s>[INST] You are an expert sentiment classifier.

    Classify the sentiment into:
    positive, negative, or neutral.

    Only output ONE word.

    Review:
    {row['review']}

    Aspect:
    {row['aspect']}

    Answer:
    [/INST] {row['sentiment']}"""
```

```
In [10]: # =====
# CREATE TEXT
# =====

train_df["text"] = train_df.apply(
    format_instruction,
    axis=1
)

val_df["text"] = val_df.apply(
    format_instruction,
    axis=1
)

test_df["text"] = test_df.apply(
    format_instruction,
    axis=1
)
```

```
In [11]: # =====
# TOKENIZER
# =====

model_name = "meta-llama/Meta-Llama-3-8B-Instruct"

tokenizer = AutoTokenizer.from_pretrained(
    model_name,
    token=True
)
```

```
tokenizer.pad_token = tokenizer.eos_token
```

```
In [12]: # =====
# TOKENIZATION
# =====

def tokenize_function(example):

    full_text = example["text"]

    prompt, answer = full_text.split("[/INST]")

    prompt = prompt + "[/INST]"

    answer = answer.strip()

    prompt_ids = tokenizer(
        prompt,
        truncation=True,
        max_length=128
    )["input_ids"]

    answer_ids = tokenizer(
        answer,
        truncation=True,
        max_length=10
    )["input_ids"]

    input_ids = prompt_ids + answer_ids

    labels = (
        [-100] * len(prompt_ids)
        + answer_ids
    )

    max_len = 128

    input_ids = input_ids[:max_len]

    labels = labels[:max_len]

    padding_length = max_len - len(input_ids)

    input_ids += (
        [tokenizer.pad_token_id]
        * padding_length
    )

    labels += (
        [-100]
        * padding_length
    )

    attention_mask = [

        1 if token != tokenizer.pad_token_id else 0

        for token in input_ids
    ]
```

```

return {

    "input_ids": input_ids,

    "attention_mask": attention_mask,

    "labels": labels,

    "class_label": example["label"]
}

```

```

In [13]: # =====
# DATASETS
# =====

train_dataset = Dataset.from_pandas(
    train_df[["text", "label"]]
)

val_dataset = Dataset.from_pandas(
    val_df[["text", "label"]]
)

test_dataset = Dataset.from_pandas(
    test_df[["text", "label"]]
)

train_dataset = train_dataset.map(
    tokenize_function,
    remove_columns=["text", "label"]
)

val_dataset = val_dataset.map(
    tokenize_function,
    remove_columns=["text", "label"]
)

test_dataset = test_dataset.map(
    tokenize_function,
    remove_columns=["text", "label"]
)

train_dataset.set_format(type="torch")

val_dataset.set_format(type="torch")

test_dataset.set_format(type="torch")

```

```

Map: 100%|██████████| 7498/7498 [00:05<00:00, 1276.24 examples/s]
Map: 100%|██████████| 938/938 [00:00<00:00, 1036.86 examples/s]
Map: 100%|██████████| 844/844 [00:00<00:00, 937.45 examples/s]

```

```

In [14]: # =====
# CUSTOM TRAINER
# =====

class WeightedTrainer(Trainer):

```

```

def compute_loss(
    self,
    model,
    inputs,
    return_outputs=False,
    **kwargs
):
    labels = inputs.pop("labels")
    class_label = inputs.pop("class_label")
    outputs = model(**inputs)
    logits = outputs.logits
    shift_logits = logits[:, :-1, :].contiguous()
    shift_labels = labels[:, 1:].contiguous()
    loss_fct = nn.CrossEntropyLoss(
        reduction="none"
    )
    loss = loss_fct(
        shift_logits.view(
            -1,
            shift_logits.size(-1)
        ),
        shift_labels.view(-1)
    )
    loss = loss.view(
        shift_labels.size()
    )
    mask = shift_labels != -100
    loss = loss * mask
    sequence_loss = (
        loss.sum(dim=1)
        / mask.sum(dim=1)
    )
    weights = class_weights.to(
        class_label.device
    )[class_label]
    sequence_loss = (

```

```

        sequence_loss * weights
    )

    loss = sequence_loss.mean()

    return (
        (loss, outputs)
        if return_outputs
        else loss
    )

```

```

In [15]: # =====
# EXPERIMENT FUNCTION
# =====
import os
def run_experiment(

    experiment_name,

    learning_rate=1e-5,

    epochs=5
):

    gc.collect()

    torch.cuda.empty_cache()

    # =====
    # CREATE EXPERIMENT FOLDER
    # =====

    experiment_dir = f"./experiments_new/{experiment_name}"

    os.makedirs(
        experiment_dir,
        exist_ok=True
    )

    # =====
    # QUANTIZATION
    # =====

    bnb_config = BitsAndBytesConfig(

        load_in_4bit=True,

        bnb_4bit_compute_dtype=torch.float16,

        bnb_4bit_use_double_quant=True,

        bnb_4bit_quant_type="nf4"
    )

    # =====
    # LOAD MODEL
    # =====

    gc.collect()

```

```

torch.cuda.empty_cache()
torch.cuda.ipc_collect()

model = AutoModelForCausalLM.from_pretrained(
    model_name,
    quantization_config=bnb_config,
    device_map="auto",
    torch_dtype=torch.float16,
    low_cpu_mem_usage=True
)

# =====
# PREPARE MODEL
# =====

model = prepare_model_for_kbit_training(
    model
)

# =====
# LORA CONFIG
# =====

lora_config = LoraConfig(

    r=8,

    lora_alpha=16,

    target_modules=[
        "q_proj",
        "v_proj"
    ],

    lora_dropout=0.05,

    bias="none",

    task_type="CAUSAL_LM"
)

# =====
# ADD LORA
# =====

model = get_peft_model(
    model,
    lora_config
)

model.config.use_cache = False

model.gradient_checkpointing_enable()

model.train()

```

```

# =====
# TRAINING ARGS
# =====

training_args = TrainingArguments(

    output_dir=experiment_dir,

    per_device_train_batch_size=1,

    gradient_accumulation_steps=8,

    num_train_epochs=epochs,

    learning_rate=learning_rate,

    logging_steps=20,

    save_strategy="epoch",

    save_total_limit=1,

    fp16=True,

    gradient_checkpointing=True,

    report_to="none",

    remove_unused_columns=False,

    optim="paged_adamw_8bit",

    max_grad_norm=1.0
)

# =====
# TRAINER
# =====

trainer = WeightedTrainer(

    model=model,

    args=training_args,

    train_dataset=train_dataset,

    eval_dataset=val_dataset
)

# =====
# TRAIN
# =====

trainer.train()

# =====

```

```

# INFERENCE FUNCTION
# =====

def predict(df):

    model.eval()

    preds = []

    labels = []

    device = next(model.parameters()).device

    for i in range(len(df)):

        review = df.iloc[i]["review"]

        aspect = df.iloc[i]["aspect"]

        true_label = df.iloc[i]["sentiment"]

        prompt = f"""<s>[INST] You are an expert sentiment classifier.

Classify the sentiment into:
positive, negative, or neutral.

Only output ONE word.

Review:
{review}

Aspect:
{aspect}

Answer:
[/INST]"""

        inputs = tokenizer(

            prompt,

            return_tensors="pt",

            truncation=True,

            max_length=128

        ).to(device)

        with torch.no_grad():

            output = model.generate(

                **inputs,

                max_new_tokens=3,

                do_sample=False,

                temperature=0.0,

```

```

        pad_token_id=tokenizer.eos_token_id
    )

    decoded = tokenizer.decode(
        output[0],
        skip_special_tokens=True
    )

    response = decoded.split(
        "[/INST]"
    )[-1].strip().lower()

    if "positive" in response:
        pred = "positive"

    elif "negative" in response:
        pred = "negative"

    elif "neutral" in response:
        pred = "neutral"

    else:
        pred = "neutral"

    preds.append(pred)

    labels.append(true_label)

    return preds, labels

# =====
# EVALUATION
# =====

print("\n==== CROSS DOMAIN RESULTS =====")

preds_cross, labels_cross = predict(test_df)

accuracy = accuracy_score(
    labels_cross,
    preds_cross
)

macro_f1 = f1_score(
    labels_cross,
    preds_cross,
    average="macro"
)

weighted_f1 = f1_score(
    labels_cross,
    preds_cross,

```

```

        average="weighted"
    )

    print("Accuracy:", accuracy)

    print("Macro F1:", macro_f1)

    print("Weighted F1:", weighted_f1)

    print("\nClassification Report:\n")

    print(
        classification_report(
            labels_cross,
            preds_cross
        )
    )

# =====
# SAVE PREDICTIONS
# =====

final_predictions_df = test_df.copy()

final_predictions_df[
    "true_sentiment"
] = labels_cross

final_predictions_df[
    "predicted_sentiment"
] = preds_cross

final_predictions_df.to_csv(

    f"{experiment_dir}/{experiment_name}_predictions.csv",

    index=False
)

# =====
# SAVE METRICS
# =====

metrics_df = pd.DataFrame({

    "Metric": [

        "Accuracy",

        "Macro_F1",

        "Weighted_F1"
    ],

    "Score": [

        accuracy,

```

```

        macro_f1,
        weighted_f1
    ]
})

metrics_df.to_csv(

    f"{experiment_dir}/{experiment_name}_metrics.csv",

    index=False
)

# =====
# SAVE MODEL
# =====

model.save_pretrained(

    f"{experiment_dir}/model"
)

tokenizer.save_pretrained(

    f"{experiment_dir}/model"
)

# =====
# CLEANUP
# =====

trainer.optimizer = None

trainer.lr_scheduler = None

del trainer

del model

gc.collect()

torch.cuda.empty_cache()

torch.cuda.ipc_collect()

return {

    "accuracy": accuracy,

    "macro_f1": macro_f1,

    "weighted_f1": weighted_f1
}

```

```

In [16]: metrics = run_experiment(

    "llama_epochs5",

)

```

```
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!  
Loading weights: 100%|██████████| 291/291 [00:09<00:00, 31.93it/s]
```

 [4690/4690 4:18:28, Epoch 5/5]

Step	Training Loss
20	774.649365
40	1012.220996
60	515.362598
80	736.585547
100	362.231958
120	839.203906
140	740.054443
160	155.096741
180	32.179572
200	10.489336
220	19.654002
240	10.390755
260	73.705060
280	11.236713
300	10.344445
320	40.662744
340	18.854935
360	26.610822
380	4.425980
400	8.312242
420	34.319083
440	14.954707
460	16.762103
480	32.890427
500	30.547504
520	22.657623
540	9.565662
560	216.390039
580	10.147418
600	17.997670
620	16.234346
640	63.682428

Step	Training Loss
660	25.338124
680	19.570439
700	62.273218
720	13.558939
740	9.536961
760	13.038116
780	19.515309
800	11.423041
820	90.590564
840	11.949101
860	16.933707
880	23.502031
900	15.572548
920	10.768750
940	3.883345
960	15.101996
980	10.989848
1000	56.512744
1020	13.947304
1040	9.077877
1060	49.942548
1080	15.902077
1100	27.768146
1120	40.736966
1140	35.310056
1160	2.786650
1180	35.571448
1200	8.402561
1220	33.056174
1240	6.320322
1260	8.512800
1280	11.931657
1300	55.649060

Step	Training Loss
1320	12.057509
1340	7.685420
1360	5.433897
1380	3.138223
1400	41.008002
1420	9.354727
1440	19.544826
1460	19.019714
1480	28.097592
1500	3.635925
1520	8.875319
1540	6.967220
1560	42.303040
1580	12.629921
1600	43.865805
1620	12.566111
1640	26.198242
1660	20.278789
1680	9.572257
1700	43.167776
1720	14.489856
1740	10.181316
1760	7.480634
1780	8.886651
1800	94.644836
1820	18.915556
1840	5.142204
1860	62.992456
1880	22.993542
1900	6.401669
1920	22.450735
1940	16.498334
1960	5.481169

Step	Training Loss
1980	13.468419
2000	28.473126
2020	27.796658
2040	12.901796
2060	8.078000
2080	24.673395
2100	93.823773
2120	17.914799
2140	7.885858
2160	4.587742
2180	7.932601
2200	8.224919
2220	6.334219
2240	37.814362
2260	19.751311
2280	7.544733
2300	17.392087
2320	12.139931
2340	7.449901
2360	6.900673
2380	17.864040
2400	15.842868
2420	77.297681
2440	83.169897
2460	5.647494
2480	13.871194
2500	28.099927
2520	12.131429
2540	6.842231
2560	65.226239
2580	6.974068
2600	19.297656
2620	6.939314

Step	Training Loss
2640	19.346376
2660	8.410976
2680	28.246881
2700	18.372459
2720	9.000500
2740	37.040063
2760	20.472012
2780	38.618179
2800	16.083444
2820	2.531398
2840	32.805038
2860	40.448831
2880	9.039983
2900	8.668587
2920	4.881153
2940	55.041937
2960	4.115124
2980	7.084827
3000	5.341616
3020	16.882259
3040	8.383945
3060	96.245935
3080	10.285970
3100	21.870183
3120	10.446986
3140	17.366183
3160	12.276665
3180	5.003854
3200	10.147192
3220	10.986712
3240	14.952864
3260	21.466975
3280	8.607915

Step	Training Loss
3300	2.481992
3320	46.159567
3340	57.495917
3360	9.107503
3380	53.175085
3400	10.359914
3420	30.220230
3440	20.170932
3460	30.539758
3480	7.610677
3500	12.949263
3520	11.438261
3540	19.937299
3560	13.097423
3580	14.955280
3600	55.737463
3620	96.245636
3640	47.872293
3660	6.234345
3680	6.634170
3700	14.624648
3720	14.110368
3740	17.826857
3760	6.395318
3780	3.448629
3800	9.395292
3820	8.119810
3840	16.892603
3860	6.212790
3880	7.967150
3900	12.928563
3920	7.683340
3940	16.479320

Step	Training Loss
3960	40.766864
3980	42.256879
4000	4.041899
4020	4.097271
4040	4.944579
4060	4.661024
4080	8.635501
4100	23.455089
4120	19.307108
4140	140.882947
4160	5.995676
4180	7.156276
4200	12.389925
4220	86.748254
4240	16.847147
4260	32.117068
4280	24.474365
4300	7.570348
4320	62.100958
4340	8.300314
4360	9.197260
4380	29.911420
4400	15.334729
4420	8.093248
4440	13.628502
4460	2.566122
4480	29.302097
4500	6.504517
4520	2.569731
4540	4.461019
4560	17.557516
4580	11.810676
4600	13.497499

Step	Training Loss
4620	24.431261
4640	44.298029
4660	22.788440
4680	171.519604

```
[transformers] The following generation flags are not valid and may be ignored:
['temperature']. Set `TRANSFORMERS_VERBOSITY=info` for more details.
[transformers] Both `max_new_tokens` (=3) and `max_length` (=4096) seem to have be
en set. `max_new_tokens` will take precedence. Please refer to the documentation
for more information. (https://huggingface.co/docs/transformers/main/en/main\_classes/text\_generation)
```

```
===== CROSS DOMAIN RESULTS =====
```

[transformers] Ignoring clean_up_tokenization_spaces=True for BPE tokenizer TokenizersBackend. The clean_up_tokenization post-processing step is designed for Word Piece tokenizers and is destructive for BPE (it strips spaces before punctuation). Set clean_up_tokenization_spaces=False to suppress this warning, or set clean_up_tokenization_spaces_for_bpe_even_though_it_will_corrupt_output=True to force cleanup anyway.

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

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for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

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[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

Accuracy: 0.773696682464455

Macro F1: 0.6537099550427153

Weighted F1: 0.7635920914244507

Classification Report:

	precision	recall	f1-score	support
negative	0.87	0.66	0.75	288
neutral	0.42	0.32	0.37	87
positive	0.78	0.93	0.85	469
accuracy			0.77	844
macro avg	0.69	0.64	0.65	844
weighted avg	0.77	0.77	0.76	844

```
In [17]: metrics = run_experiment(
    "llama_epochs3",
    epochs=3
)
```

Loading weights: 100%|██████████| 291/291 [00:08<00:00, 33.40it/s]

 [2814/2814 2:49:37, Epoch 3/3]

Step	Training Loss
20	774.525098
40	1009.564648
60	504.717920
80	712.047852
100	337.570215
120	800.669580
140	672.510791
160	147.045776
180	29.772437
200	10.749249
220	19.662469
240	10.548779
260	70.418494
280	11.354215
300	10.363589
320	38.208932
340	18.502394
360	27.288644
380	4.340562
400	8.513641
420	34.547531
440	14.521747
460	17.385538
480	33.521954
500	30.941684
520	22.632793
540	10.308419
560	214.742847
580	10.264623
600	17.620026
620	16.705553
640	66.113318

Step	Training Loss
660	25.576450
680	19.924300
700	52.416779
720	13.939058
740	9.641662
760	13.005385
780	17.759708
800	11.062339
820	88.251343
840	11.201078
860	15.699792
880	23.963446
900	15.211079
920	10.837350
940	4.026986
960	14.989587
980	11.066652
1000	57.105676
1020	15.108180
1040	9.264664
1060	50.027539
1080	16.717651
1100	28.501443
1120	40.605246
1140	37.954459
1160	3.079951
1180	36.779837
1200	8.610495
1220	32.991028
1240	7.532897
1260	9.957346
1280	15.511732
1300	49.263107

Step	Training Loss
1320	13.365695
1340	8.849575
1360	5.610838
1380	3.389900
1400	41.776181
1420	8.438934
1440	19.163882
1460	21.948544
1480	29.025922
1500	3.802719
1520	8.625494
1540	6.578704
1560	43.262027
1580	13.084514
1600	58.089276
1620	14.692781
1640	28.371057
1660	21.597652
1680	11.830654
1700	45.530759
1720	15.518863
1740	11.612654
1760	6.995366
1780	9.116880
1800	95.760919
1820	24.324954
1840	5.319046
1860	65.076654
1880	20.865193
1900	6.860561
1920	26.835913
1940	14.744604
1960	5.765668

Step	Training Loss
1980	14.191089
2000	27.027426
2020	38.696304
2040	12.997685
2060	6.953418
2080	23.589081
2100	78.289105
2120	19.754446
2140	7.013208
2160	5.094138
2180	6.855659
2200	10.337881
2220	8.167151
2240	43.533554
2260	17.969138
2280	8.525860
2300	18.972697
2320	13.271994
2340	8.685862
2360	7.644510
2380	18.622441
2400	13.243442
2420	64.364404
2440	78.003973
2460	6.930399
2480	16.533472
2500	26.504639
2520	11.030931
2540	7.433385
2560	53.334784
2580	8.718659
2600	26.337357
2620	9.746516

Step	Training Loss
2640	16.619534
2660	6.238309
2680	25.315340
2700	19.196416
2720	11.127569
2740	30.351801
2760	14.393306
2780	23.586263
2800	13.228590

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

===== CROSS DOMAIN RESULTS =====

[illegible]

[illegible]

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[illegible]

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```
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```

[illegible]

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Accuracy: 0.7654028436018957

Macro F1: 0.6349833848193615

Weighted F1: 0.7546779503728467

Classification Report:

	precision	recall	f1-score	support
negative	0.85	0.66	0.75	288
neutral	0.38	0.28	0.32	87
positive	0.78	0.92	0.84	469
accuracy			0.77	844
macro avg	0.67	0.62	0.63	844
weighted avg	0.76	0.77	0.75	844

```
In [ ]: metrics = run_experiment(
    "llama_lr3e-5",
    learning_rate=3e-5
)
```

Loading weights: 100%|██████████| 291/291 [00:09<00:00, 31.58it/s]

 [4310/4690 4:31:55 < 23:59, 0.26 it/s,

Epoch 4.59/5]

Step	Training Loss
20	771.495703
40	925.600000
60	339.801953
80	180.297400
100	12.978755
120	44.411905
140	96.716919
160	66.019641
180	30.449985
200	5.984056
220	20.461285
240	7.473244
260	79.935986
280	9.402651
300	9.312331
320	61.708148
340	16.975531
360	26.062613
380	4.081044
400	8.338871
420	36.010013
440	20.781145
460	14.313712
480	18.149139
500	34.703024
520	23.082332
540	5.882409
560	148.195984
580	7.850050
600	17.040820
620	12.867566

Step	Training Loss
640	67.150568
660	23.989143
680	18.664058
700	141.060876
720	12.160101
740	9.294749
760	12.812019
780	20.099158
800	9.936813
820	96.709955
840	17.159433
860	20.288503
880	16.049123
900	16.367140
920	7.343010
940	2.207703
960	12.764280
980	8.034630
1000	63.391687
1020	4.265579
1040	8.745074
1060	50.060748
1080	16.036983
1100	23.839790
1120	62.565686
1140	31.362872
1160	2.811581
1180	40.382092
1200	9.223119
1220	40.363431
1240	9.556074
1260	7.865966
1280	5.312523

Step	Training Loss
1300	52.136896
1320	9.788607
1340	4.414210
1360	5.426986
1380	3.276999
1400	33.509225
1420	8.874746
1440	18.145728
1460	14.288042
1480	20.739301
1500	3.161607
1520	9.154284
1540	6.224710
1560	27.186777
1580	11.425500
1600	16.232350
1620	9.417205
1640	37.256998
1660	13.557907
1680	7.332099
1700	31.634195
1720	13.401576
1740	6.664864
1760	6.355330
1780	8.187277
1800	62.270703
1820	10.875796
1840	3.407473
1860	42.059927
1880	21.225099
1900	5.042448
1920	20.331508
1940	20.004984

Step	Training Loss
1960	6.548607
1980	6.170763
2000	58.611603
2020	29.847443
2040	13.105231
2060	4.551534
2080	40.604227
2100	128.177405
2120	16.178983
2140	4.051928
2160	4.902665
2180	6.195447
2200	7.021870
2220	5.436994
2240	12.016093
2260	18.325032
2280	6.445975
2300	15.178142
2320	7.926240
2340	4.067942
2360	9.153983
2380	29.442169
2400	8.396793
2420	56.939832
2440	80.924701
2460	2.738693
2480	5.015585
2500	30.316238
2520	9.715270
2540	5.217382
2560	66.078925
2580	3.994310
2600	8.308787

Step	Training Loss
2620	7.834705
2640	22.060666
2660	7.273526
2680	33.255161
2700	8.013409
2720	5.020630
2740	102.489832
2760	30.104810
2780	34.598993
2800	19.325807
2820	1.923758
2840	39.994998
2860	43.460013
2880	7.428768
2900	3.315413
2920	3.657026
2940	35.940485
2960	3.102958
2980	4.084430
3000	5.681909
3020	13.175275
3040	5.255923
3060	17.398584
3080	5.598344
3100	17.285706
3120	5.897627
3140	11.707778
3160	1.766486
3180	5.565001
3200	5.926265
3220	7.668806
3240	2.445119
3260	8.376172

Step	Training Loss
3280	5.016521
3300	2.029897
3320	63.776813
3340	40.735858
3360	7.676367
3380	37.939490
3400	8.286077
3420	33.247015
3440	11.328899
3460	33.558829
3480	4.772568
3500	4.447989
3520	6.480640
3540	12.801085
3560	3.119044
3580	14.415382
3600	79.888812
3620	103.748608
3640	7.252425
3660	4.443478
3680	3.539445
3700	14.836981
3720	10.252030
3740	14.296902
3760	6.626544
3780	2.721676
3800	4.690338
3820	2.149296
3840	6.867613
3860	3.265918
3880	5.180239
3900	2.546186
3920	0.156355

Step	Training Loss
3940	24.830180
3960	37.151367
3980	10.514220
4000	2.230220
4020	0.671428
4040	1.745722
4060	3.672891
4080	1.548113
4100	4.046560
4120	11.882740
4140	74.635828
4160	1.044800
4180	6.754353
4200	12.271457
4220	10.219105
4240	23.634100
4260	4.252515
4280	16.204123
4300	1.272142

In []: